

Diabetic Retinopathy — Medical Background for SIH26038

1. What is Diabetic Retinopathy?

Diabetic Retinopathy (DR) is an eye disease caused by long-term damage to the small blood vessels in the **retina** due to diabetes.

The **retina** is the light-sensitive layer at the back of the eye. It receives light and sends visual information to the brain through the optic nerve.

When blood glucose remains high over time, the tiny blood vessels supplying the retina can become damaged. They may become blocked, leak fluid or blood, and, in advanced disease, trigger the growth of abnormal new blood vessels.

The important point for screening is:

Diabetic retinopathy may cause little or no noticeable symptoms in its early stages.

Therefore, a person can have retinal damage while their vision still seems normal. Early detection and regular eye examinations are important because timely management can reduce the risk of serious vision loss.

2. Understanding the Retina in Simple Terms

Think of the eye as a camera.

```
graph TD; Light --> Cornea; Cornea --> Lens; Lens --> Retina; Retina --> Optic_Nerve[Optic Nerve]; Optic_Nerve --> Brain;
```

The **retina** works somewhat like the camera sensor.

It contains very small blood vessels because retinal cells need oxygen and nutrients continuously.

These vessels are the structures that become damaged in diabetic retinopathy.

Important retinal structures

Structure	Simple explanation	Importance to SIH26038
Retina	Light-sensitive tissue at the back of the eye	Main area analyzed in fundus images
Macula	Central retinal region responsible for detailed central vision	Important for DME assessment
Fovea	Small central region providing the sharpest vision	Localization target
Optic Disc	Area where the optic nerve exits the eye	Localization/segmentation target
Blood Vessels	Supply oxygen and nutrients to retina	Vessel segmentation

IDRiD specifically provides optic-disc and fovea location information, while SIH26038 requires analysis of retinal structures and lesions.

3. Why Does Diabetes Damage the Retina?

The basic chain is:

```
Diabetes
↓
Long-term high blood glucose
↓
Damage to tiny retinal blood vessels
↓
Leakage and/or blockage
↓
Retinal damage and reduced blood supply
↓
Diabetic Retinopathy
```

Damaged vessels can become leaky or blocked. When areas of the retina receive insufficient blood supply, the retina can respond by releasing signals that promote abnormal new blood-vessel growth.

This process can therefore be simplified into two major problems:

Problem A — Leakage

Blood vessels become damaged and leak fluid or blood.

This can cause retinal swelling and lesions visible in a fundus photograph.

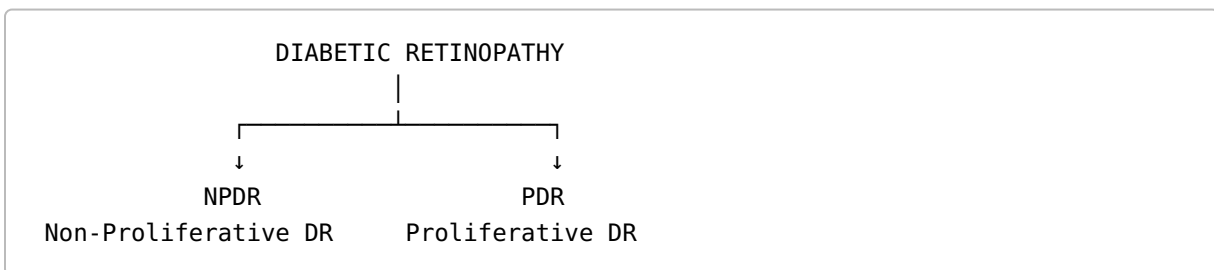
Problem B — Ischemia / inadequate blood supply

Some retinal vessels become blocked, reducing oxygen delivery.

In advanced disease, this can stimulate abnormal new blood vessels.

4. Main Types of Diabetic Retinopathy

Clinically, diabetic retinopathy is broadly divided into two major forms:



The defining difference is **neovascularization**, meaning the growth of abnormal new blood vessels.

5. Type 1 — Non-Proliferative Diabetic Retinopathy (NPDR)

NPDR is generally the earlier form of diabetic retinopathy.

In NPDR, the retinal blood vessels are damaged, but **abnormal new blood vessels have not yet developed**.

NPDR is commonly divided into:

1. Mild NPDR
2. Moderate NPDR
3. Severe NPDR

These stages reflect increasing retinal vascular abnormalities.

5.1 Mild NPDR

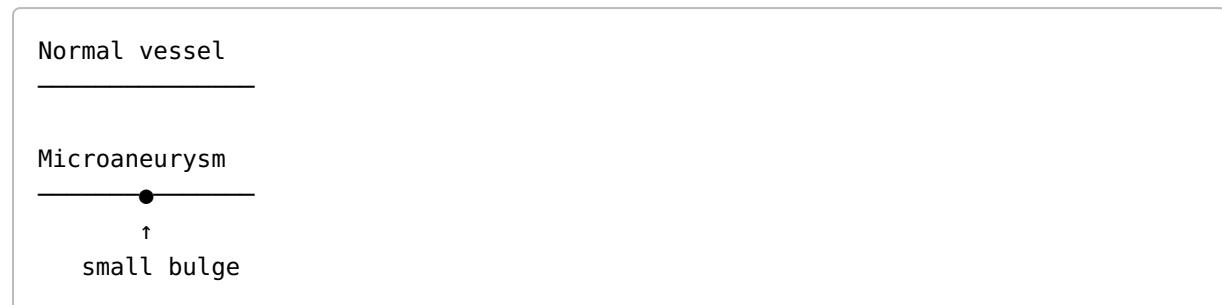
This is the earliest recognizable stage.

The typical finding is:

Microaneurysms

A microaneurysm is a small bulging/outpouching of a retinal blood vessel.

Think of it as a tiny weak spot in a small blood vessel.



These may leak fluid.

AI relevance

Microaneurysms are important targets for computer-vision analysis because they can appear as tiny abnormal regions in fundus photographs.

IDRiD provides pixel-level annotations for microaneurysms in its lesion-annotated subset.

6. Moderate NPDR

As retinal vascular damage increases, more abnormalities can appear.

Possible findings include:

- More microaneurysms
- Retinal hemorrhages
- Hard exudates
- Cotton-wool spots
- Other vascular abnormalities

The exact clinical assessment is based on the overall retinal findings rather than simply counting one lesion type.

AI relevance

This stage is particularly important because the model needs to distinguish more than simply:

Healthy vs Diseased.

It should learn patterns associated with increasing severity.

7. Severe NPDR

Severe NPDR means substantial retinal vascular damage.

Many retinal blood vessels may become blocked, depriving parts of the retina of oxygen.

Clinical evaluation considers findings such as:

- Extensive retinal hemorrhages/microaneurysms
- Venous beading
- Intraretinal microvascular abnormalities (IRMA)

A commonly used clinical framework is the **4-2-1 rule** for severe NPDR.

For a non-medical understanding:

The retina is showing widespread and serious blood-vessel damage and is at high risk of progressing to proliferative disease.

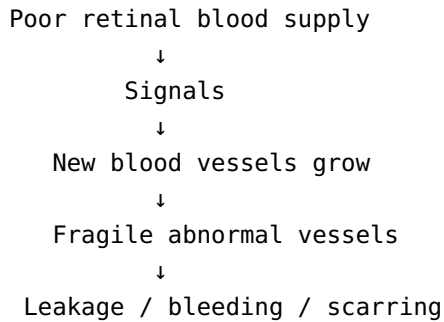
8. Type 2 — Proliferative Diabetic Retinopathy (PDR)

PDR is the advanced form of diabetic retinopathy.

When areas of the retina do not receive enough oxygen, the retina can release signals that stimulate the formation of new blood vessels.

This is called:

Neovascularization



These abnormal vessels are fragile and can bleed. Scar tissue associated with advanced disease can also pull on the retina and cause a tractional retinal detachment.

AI relevance

SIH26038 specifically mentions detection/analysis of **neovascularization** as an important retinal abnormality.

Therefore, PDR should be treated as a major high-risk finding in the screening workflow.

9. The Five-Level Classification Relevant to Our AI

For SIH26038, the disease can be represented using the **International Clinical Diabetic Retinopathy Severity Scale**:

AI Level	Clinical category	Simple meaning
0	No apparent DR	No visible DR signs
1	Mild NPDR	Early retinal blood-vessel changes
2	Moderate NPDR	More extensive retinal abnormalities
3	Severe NPDR	Extensive vascular damage; high risk of progression
4	Proliferative DR	Abnormal new blood vessels

The five-stage scale was developed to provide consistent clinical classification and communication.

This is directly relevant to our **5-class AI classifier**.

10. Important Clarification: DME Is Not Simply Another DR Stage

A common misconception is to treat **Diabetic Macular Edema (DME)** as Level 5.

It is not.

DME is a **complication that can occur at different stages of diabetic retinopathy**.

DME occurs when damaged retinal blood vessels leak fluid into/around the macula, causing retinal swelling.

Because the macula is responsible for detailed central vision, DME can significantly affect vision.

Diabetic Retinopathy



DME can occur alongside DR at different stages

This distinction is important for both our medical explanation and our dataset design.

11. Important Retinal Lesions

The major lesions relevant to our computer-vision system include:

11.1 Microaneurysms

Small abnormal bulges in retinal capillaries.

Appearance: Tiny dots.

Importance: Early sign of DR.

AI task: Detection/segmentation.

11.2 Hemorrhages

Areas where blood has escaped from damaged retinal vessels.

Appearance: Dark/red retinal spots or blot-like regions depending on location and appearance.

Importance: Can indicate increasing disease severity.

AI task: Detection/segmentation.

11.3 Hard Exudates

Deposits of lipid/protein material resulting from leakage from damaged blood vessels.

Appearance: Often bright/yellow-white regions.

Importance: Indicates retinal vascular leakage and can be particularly important around the macula.

AI task: Detection/segmentation.

11.4 Soft Exudates / Cotton-Wool Spots

Areas associated with localized retinal ischemic damage.

Appearance: Pale/white fluffy-looking retinal patches.

Importance: Can occur with more significant retinal vascular disease.

AI task: Detection/segmentation.

11.5 Venous Beading

Retinal veins can develop irregular changes in their appearance.

Importance: One of the findings considered when determining severe NPDR.

AI relevance: More difficult than detecting simple dots or bright lesions; potentially useful as an advanced feature rather than an initial MVP target.

11.6 IRMA

Intraretinal Microvascular Abnormalities (IRMA) are abnormal retinal vascular changes occurring within the retina.

They are clinically important in severe NPDR.

AI relevance: Advanced lesion/vascular-feature analysis.

11.7 Neovascularization

Abnormal new blood-vessel growth.

Importance: Key feature of PDR.

AI relevance: Important advanced-stage indicator and potential segmentation/detection target.

12. Main Complications of Diabetic Retinopathy

Untreated or advanced diabetic retinopathy can lead to several serious complications.

12.1 Diabetic Macular Edema (DME)

Damaged vessels leak fluid, causing swelling in the macula.

```
graph TD; A[Damaged vessel] --> B[Fluid leakage]; B --> C[Macular swelling]; C --> D[Blurred central vision];
```

DME can occur at different stages of DR.

SIH relevance

The model should identify retinal findings associated with DME risk, while recognizing that a fundus-photo screening system is not a substitute for tests such as OCT when detailed macular assessment is clinically required.

13. Vitreous Hemorrhage

Fragile abnormal blood vessels in advanced disease can bleed into the vitreous, the gel-like material inside the eye.

This can cause:

- Floaters
- Dark spots
- Marked reduction in vision

Significant bleeding requires prompt ophthalmic assessment.

SIH relevance

The screening report should flag advanced/high-risk findings rather than treating an AI result as a final diagnosis.

14. Retinal Detachment

Advanced disease can produce scar tissue.

When scar tissue contracts, it can pull the retina away from its normal position.

This is called **tractional retinal detachment**.

It can cause severe and potentially permanent vision loss.

SIH relevance

This demonstrates why detecting advanced disease early matters.

15. Neovascular Glaucoma

Abnormal blood vessels associated with advanced diabetic retinopathy can interfere with the normal drainage of fluid from the eye and may contribute to a severe form of glaucoma.

This can cause substantial vision loss.

SIH relevance

PDR detection is therefore not merely about classification accuracy; identifying advanced disease can support appropriate referral.

16. Symptoms

One of the biggest challenges with DR is that **symptoms may appear late**.

Early disease can have no obvious symptoms.

Possible later symptoms include:

- Blurred vision
- Difficulty seeing clearly
- Floaters or dark spots
- Areas of reduced vision
- Severe vision loss in advanced disease

The National Eye Institute emphasizes that people may have DR before noticing symptoms.

Why this matters for our project

A patient should not be told:

"My vision is normal, so my retina must be normal."

Instead:

Screening should help detect retinal disease before significant symptoms or vision loss occur.

This is one of the strongest arguments for a screening-oriented system.

17. Risk Factors

The most important underlying factor is diabetes itself, but the risk of diabetic retinopathy increases with factors such as:

- Longer duration of diabetes
- Poor blood-glucose control
- High blood pressure
- High cholesterol

- Pregnancy in people with diabetes
- Other individual clinical risk factors

The CDC and ophthalmology guidance emphasize diabetes duration and control of glucose and blood pressure among important risk factors.

18. Prevention and Risk Reduction

DR cannot be addressed by an eye-screening model alone.

The broader prevention strategy includes:

```
Diabetes management
      +
Blood-pressure management
      +
Regular eye screening
      +
Timely ophthalmology care
      ↓
Lower risk of serious vision loss
```

Healthy lifestyle habits and appropriate diabetes treatment are important parts of reducing complications.

Important project boundary

Our SIH system should **not prescribe medicines or independently decide treatment**.

Its role should be:

Screen → Detect → Explain → Risk-stratify → Refer

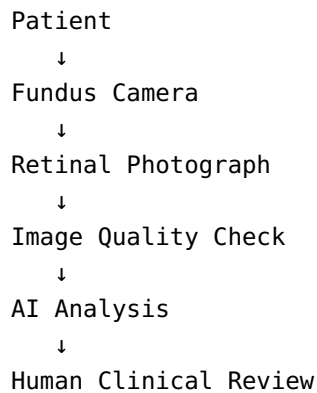
rather than:

Diagnose → Prescribe → Treat

19. How Is Diabetic Retinopathy Detected?

A common screening approach is **retinal/fundus photography**.

A fundus camera captures an image of the back of the eye.



Regular eye examinations are important because early disease can be asymptomatic.

Indian DR screening guidance also recognizes retinal imaging, tele-ophthalmology and potential AI grading as components of screening systems, while emphasizing referral pathways and quality assurance.

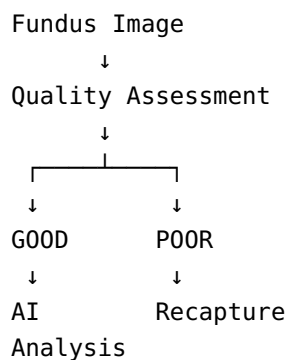
20. Why Image Quality Is Critical

An AI model cannot reliably analyze a retinal image that is:

- Too blurry
- Poorly illuminated
- Incomplete
- Poorly centred
- Severely affected by artifacts

Therefore, SIH26038's image-quality component is not just a cosmetic feature.

It is a **safety mechanism**.



The system should be able to say:

"Image not suitable for reliable screening. Please capture another image."

instead of forcing an unreliable prediction.

21. What Should Our AI Detect?

For SIH26038, the medical concepts translate into computer-vision targets.

Medical finding	AI task
Microaneurysms	Detection / segmentation
Hemorrhages	Detection / segmentation
Hard exudates	Detection / segmentation
Soft exudates	Detection / segmentation
Blood vessels	Segmentation
Optic disc	Localization / segmentation
Fovea	Localization
Neovascularization	Detection / advanced segmentation
DR severity	5-class classification
Image quality	Quality assessment
Important decision regions	Explainable AI

IDRiD is particularly useful here because its annotations include multiple DR lesions plus optic-disc and fovea information.

22. From Medical Signs to AI Prediction

Our system should not behave like a black box:

Image
↓
AI
↓
"DR detected"

Instead:

```
Image
↓
Quality Assessment
↓
Retinal Structure Analysis
↓
Lesion Detection
↓
DR Severity Classification
↓
Explainable AI
↓
Confidence
↓
Screening Report
```

For example:

```
DR Grade: Level 2
Category: Moderate NPDR

Detected evidence:
• Microaneurysm-like lesions
• Retinal hemorrhagic features
• Exudative features

AI attention:
[Grad-CAM / lesion visualization]

Recommendation:
Ophthalmology review according to
screening/referral protocol.
```

The evidence shown should be presented as **AI-supported findings**, not as a replacement for clinical diagnosis.

23. Explainable AI: Why Did the Model Say This?

Medical AI needs more than an answer.

A clinician may want to know:

Which part of the retinal image influenced the prediction?

For our project, **Grad-CAM** can generate an attention map.

```
Original Fundus Image
  ↓
  AI Model
  ↓
  Prediction
  ↓
  Grad-CAM
  ↓
Highlighted important regions
```

A stronger research approach is:

```
Grad-CAM
  ↓
Compare highlighted region
  ↓
Clinically annotated lesion mask
  ↓
Measure overlap
```

IDRiD's pixel-level lesion masks allow this type of quantitative evaluation for its annotated subset.

24. Human-in-the-Loop Screening

Our AI should **assist healthcare workers and ophthalmologists**, not replace them.

The proposed workflow is:

```
Patient
  ↓
Fundus Image
  ↓
AI Screening
  ↓
Risk / DR Grade / Explanation
  ↓
Ophthalmologist Review
  ↓
Clinical Decision
```

This is especially relevant in India, where screening guidance emphasizes referral pathways, quality assurance, and the potential role of technology/AI in grading.

25. What Does "Referable" Mean?

For a screening system, an important goal is to identify people who need further professional assessment.

The exact referral threshold depends on the screening protocol and the clinical context.

For our SIH prototype, the system should therefore provide:

```
DR Grade
  +
DME-related evidence if available
  +
Image quality
  +
Confidence
  +
Clinical referral flag
```

The referral flag should be designed according to the selected clinical guideline/protocol and reviewed by qualified clinicians.

The Indian consensus statement specifically describes the purpose of DR screening as identifying people with sight-threatening DR so they can receive timely treatment and emphasizes dynamic referral pathways.

26. Why Rural India Is an Important Target

A rural screening program may face challenges such as:

- Limited access to ophthalmologists
- Large patient volumes
- Long travel distances for specialist care
- Limited specialist availability
- Need for efficient referral

Tele-ophthalmology can help by using retinal images for remote interpretation, while AI can potentially assist with grading and prioritization. Indian DR screening guidance discusses community-based screening, tele-ophthalmology and AI-enabled grading as relevant approaches.

This is why our project is not simply:

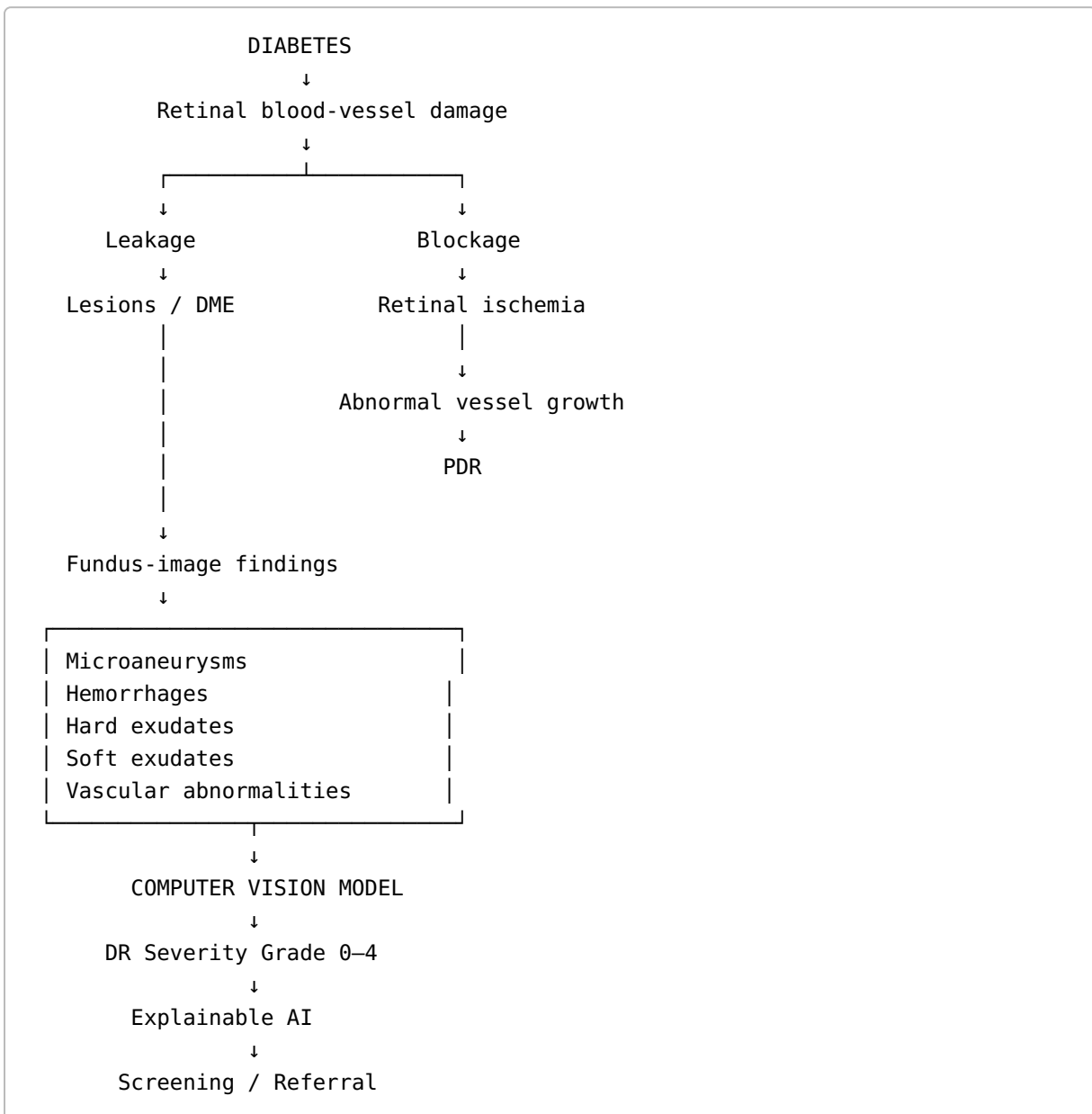
"An image classifier."

It is intended as:

"An AI-assisted screening and referral-support system."

27. Complete Medical-to-AI Mapping

This is the key section for our SIH implementation.



28. Dataset Connection

The medical requirements determine why multiple datasets are needed.

APTOS 2019

Used primarily for:

DR severity classification

Fundus image
↓
Grade 0–4

IDRiD

Used for:

Indian retinal images + lesion annotations + optic disc/fovea information

Fundus image
↓
Lesion / structure analysis

IDRiD contains 516 images from an eye clinic in Nanded, Maharashtra, India, and provides image-level DR/DME grading plus pixel-level annotations for several lesions in a subset.

DRIVE

Used for:

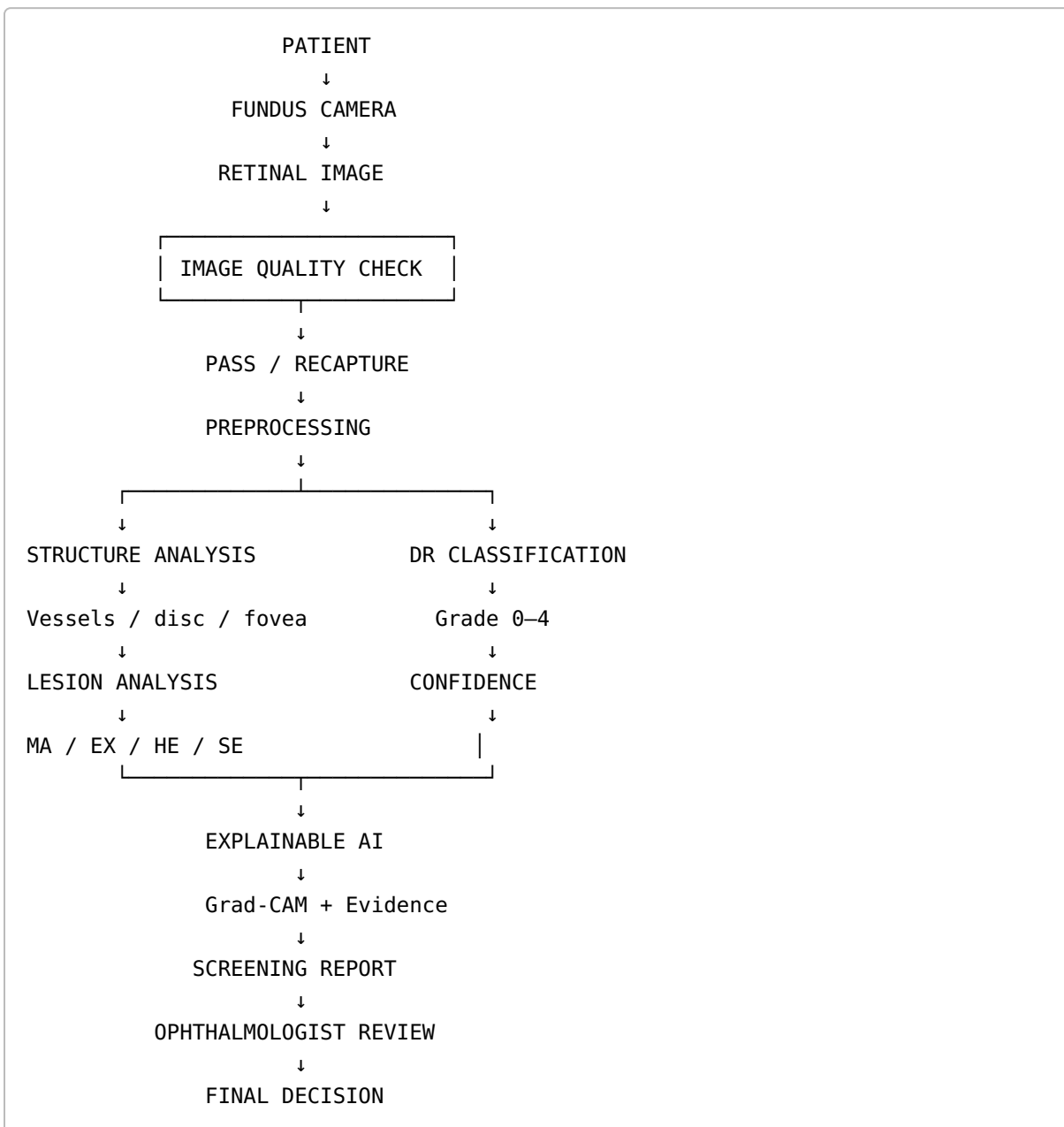
Retinal blood-vessel segmentation

Messidor-2

Used for:

External validation/generalization

29. Proposed SIH26038 Medical Pipeline



30. What the Patient Should Receive

The system should present information in understandable language.

Example:

RETINAL SCREENING REPORT

Image Quality:

GOOD ✓

AI Screening Result:

DR Grade: Level 2

Category: Moderate NPDR

Detected retinal findings:

- Microaneurysm-like lesions
- Hemorrhagic features
- Exudative features

AI Confidence:

XX%

Explanation:

Highlighted regions show areas that contributed to the AI prediction.

Action:

Further ophthalmic evaluation is advised according to the applicable screening protocol.

IMPORTANT:

This is an AI-assisted screening result, not a final medical diagnosis.

31. What Our System Must NOT Claim

To keep the project medically responsible, we should avoid claims such as:

- ✗ "The AI diagnoses every eye disease."
- ✗ "The AI replaces an ophthalmologist."
- ✗ "A normal AI result guarantees that the patient has no eye disease."
- ✗ "The AI-prescribed treatment is correct."

Instead:

- ✓ "AI-assisted diabetic retinopathy screening."
 - ✓ "Automated retinal-image quality assessment."
 - ✓ "DR severity classification."
 - ✓ "Lesion and structural evidence."
 - ✓ "Explainable prediction."
 - ✓ "Referral-support information."
 - ✓ "Ophthalmologist-in-the-loop."
-

32. Why Early Detection Matters

The major motivation for screening is simple:

```
No / mild visible disease
      ↓
Early detection
      ↓
Timely monitoring / treatment
      ↓
Lower risk of severe vision loss
```

DR can progress before a person notices symptoms, which makes regular screening particularly important.

Indian screening guidance similarly emphasizes identifying sight-threatening disease early enough for timely care.

33. Medical Summary for Non-Medical Team Members

In one simple story:

Diabetes can gradually damage the tiny blood vessels of the retina.

At first, the damage may be small and the person may not notice anything.

Small vascular abnormalities such as microaneurysms can appear.

As damage increases, hemorrhages, exudates and other vascular abnormalities may develop.

In severe disease, much of the retina may receive inadequate blood supply.

The retina can then stimulate the growth of abnormal new blood vessels.

Those vessels are fragile and may bleed or contribute to other serious complications.

Therefore, the goal of screening is to detect retinal changes **before irreversible vision loss occurs**, and to make sure people who need specialist assessment are referred appropriately.

34. Final Medical Requirements for SIH26038

Based on the medical background, our solution should focus on:

Medical requirement	System requirement
Early DR can be asymptomatic	Screening using retinal images
Image quality affects reliability	Quality-assessment gate
DR has severity levels	5-class Grade 0–4 classifier
Microaneurysms matter	Lesion detection/segmentation
Hemorrhages matter	Lesion detection/segmentation
Exudates matter	Lesion detection/segmentation
Blood vessels are important	Vessel segmentation
Optic disc/fovea are important landmarks	Localization
PDR involves neovascularization	Advanced-stage detection
DME can cause visual impairment	DME-related evidence/reporting
AI must be interpretable	Grad-CAM + lesion evidence
AI confidence matters	Calibrated confidence
Rural specialist access can be limited	Telemedicine-compatible workflow
Screening requires referral	Human-in-the-loop referral pathway
External validity matters	Independent validation

35. Core Message of Our SIH Project

Diabetic retinopathy is a progressive diabetes-related retinal vascular disease that can remain asymptomatic in its early stages. By analyzing fundus photographs, an AI-assisted screening system can help assess image quality, identify retinal lesions and structures, estimate DR severity, explain the basis of its prediction, and prioritize patients for appropriate ophthalmic evaluation.

The goal is **not to replace doctors**.

The goal is to help extend reliable screening to settings where specialist access may be limited.

References

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Project Disclaimer

This document is intended for **research and educational purposes** as part of the SIH26038 project. It does not provide individual medical diagnosis or treatment advice. A screening AI system should be clinically validated and used with appropriate professional oversight before real-world deployment.